

Midterm 2

Macroeconomics II (Spring 2026)
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Instructions

- 40 multiple-choice questions. Exactly one option is correct for each question.
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Problem 1 (Questions 1–20): Building blocks and key derivations

- ① Three elements. The IS–MP–PC model combines: _____
- (A) AD curve, AS curve, and a money demand curve
 - (B) IS curve, LM curve, and a Phillips curve
 - (C) Phillips curve, IS curve, and a monetary policy rule
 - (D) Phillips curve, money supply rule, and a government budget constraint
- ② Phillips curve (form). The baseline Phillips curve is: _____
- (A) $\pi_t = \pi_t^e + \gamma(y_t - y_t^*) + \varepsilon_t^\pi$
 - (B) $\pi_t = \pi_t^e - \gamma(y_t - y_t^*) + \varepsilon_t^\pi$
 - (C) $\pi_t = \pi^* + \gamma(y_t - y_t^*)$
 - (D) $\pi_t = \pi_{t-1} + \gamma(y_t - y_t^*)$
- ③ Interpretation of π_t^e . In the Phillips curve, π_t^e represents: _____
- (A) realized inflation last period
 - (B) expected inflation built into wage/price setting at time t

- (C) the inflation target set by the central bank
- (D) the nominal interest rate

④ **Transmission mechanism (review emphasis).** A typical chain from inflation to real activity under an inflation-targeting rule is: _____

- (A) $\pi \uparrow \Rightarrow u \downarrow \Rightarrow x \downarrow$
- (B) $\pi \uparrow \Rightarrow r \downarrow \Rightarrow x \downarrow$
- (C) $\pi \uparrow \Rightarrow y^* \uparrow \Rightarrow x \uparrow$
- (D) $\pi \uparrow \Rightarrow \text{tighter policy} \Rightarrow u \uparrow \Rightarrow x \downarrow$

⑤ **Okun connection.** If $u_t - u_t^n \approx -\lambda(y_t - y_t^*)$ with $\lambda > 0$, then $y_t > y_t^*$ implies: _____

- (A) $u_t < u_t^n$
- (B) $u_t > u_t^n$
- (C) $u_t = u_t^n$
- (D) no implication for unemployment

⑥ **IS curve (form).** The IS curve used in the notes is: _____

- (A) $y_t = y_t^* + \alpha(i_t - \pi_t - r^*) + \varepsilon_t^y$
- (B) $y_t = y_t^* - \alpha(i_t + \pi_t - r^*) + \varepsilon_t^y$
- (C) $y_t = y_t^* - \alpha(i_t - \pi_t - r^*) + \varepsilon_t^y$
- (D) $y_t = r^* - \alpha(i_t - \pi_t - y_t^*) + \varepsilon_t^y$

⑦ **What does α measure?** In the IS curve, a larger α means: _____

- (A) output responds less to the real interest rate
- (B) output responds more to the real interest rate
- (C) inflation responds more to output
- (D) the central bank responds more to inflation

- ⑧ **Monetary policy rule (baseline).** The baseline MP rule is: _____
- (A) $i_t = r^* + \pi^* - \beta_\pi(\pi_t - \pi^*)$
 - (B) $i_t = \pi_t + \beta_\pi(\pi_t - \pi^*)$
 - (C) $i_t = r^* + \pi_t + \beta_\pi(\pi_t - \pi^*)$
 - (D) $i_t = r^* + \pi^* + \beta_\pi(\pi_t - \pi^*)$
- ⑨ **What does β_π do?** A higher β_π implies: _____
- (A) a stronger increase in i_t when inflation rises above target
 - (B) a weaker increase in i_t when inflation rises above target
 - (C) a steeper Phillips curve
 - (D) a higher natural output y_t^*
- ⑩ **Special case $\beta_\pi = 1$.** Under the baseline MP rule, if $\beta_\pi = 1$ then: _____
- (A) the nominal rate does not react to inflation at all
 - (B) the real rate rises when inflation rises
 - (C) the real rate is pinned at r^* regardless of $\pi_t - \pi^*$
 - (D) the IS–MP curve is downward sloping
- ⑪ **Derivation step (review emphasis).** The IS–MP curve is obtained by: _____
- (A) substituting the Phillips curve into the MP rule
 - (B) substituting the MP rule into the IS curve
 - (C) substituting money demand into the IS curve
 - (D) substituting the IS curve into the Phillips curve
- ⑫ **IS–MP curve (result).** The derived IS–MP curve is: _____

- (A) $y_t = y_t^* - \alpha(\beta_\pi - 1)(\pi_t - \pi^*) + \varepsilon_t^y$
- (B) $y_t = y_t^* + \alpha(\beta_\pi - 1)(\pi_t - \pi^*) + \varepsilon_t^y$
- (C) $y_t = y_t^* - \alpha\beta_\pi(\pi_t - \pi^*) + \varepsilon_t^y$
- (D) $y_t = y_t^* - \alpha(\pi_t - \pi^*) + \varepsilon_t^\pi$

⑬ **Slope sign.** In a graph with output on the horizontal axis and inflation on the vertical axis, the IS–MP curve slopes down when: _____

- (A) $\beta_\pi < 1$
- (B) $\beta_\pi = 0$
- (C) $\beta_\pi < 0$
- (D) $\beta_\pi > 1$

⑭ **Why $\beta_\pi > 1$ matters (core intuition).** The Taylor principle is central because when inflation rises it should: _____

- (A) lower the real interest rate, boosting demand
- (B) raise the real interest rate, reducing demand
- (C) raise natural output y^*
- (D) make the Phillips curve flat

⑮ **PC shift: higher expected inflation.** Holding $y_t - y_t^*$ fixed, an increase in π_t^e causes: _____

- (A) π_t to rise one-for-one (PC shifts up)
- (B) π_t to fall one-for-one (PC shifts down)
- (C) no change in π_t
- (D) the slope γ to fall

⑯ **After π^e rises (qualitative result).** If π_t^e rises above the target π^* , then in the one-period IS–MP–PC intersection (with $\beta_\pi > 1$) we typically get: _____

- (A) higher output and higher inflation, and inflation exceeds expectations
- (B) lower output and lower inflation
- (C) lower output and higher inflation, but inflation rises less than expectations
- (D) higher output and lower inflation

⑪ **Soft vs tough central bank (definition).** In these notes, a “tough” central bank corresponds to: _____

- (A) smaller β_π
- (B) larger β_π
- (C) smaller γ
- (D) larger ε_t^π

⑫ **Tough CB (one-period tradeoff).** When β_π is higher (tougher CB), a cost-push increase in inflation tends to produce: _____

- (A) a larger rise in inflation and a smaller fall in output
- (B) a larger rise in inflation and a larger fall in output
- (C) a smaller rise in inflation and a smaller fall in output
- (D) a smaller rise in inflation and a larger fall in output

⑬ **Numerical slope check.** Suppose $\alpha = 2$ and $\beta_\pi = 2$. The coefficient on $(\pi_t - \pi^*)$ in the IS–MP curve is: _____

- (A) -2
- (B) -4
- (C) $+2$
- (D) 0

⑭ **Interpretation.** In the IS–MP curve, a positive ε_t^y shifts the curve: _____

- (A) left (lower y for each π)
 - (B) right (higher y for each π)
 - (C) up (higher π for each y)
 - (D) down (lower π for each y)
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Problem 2 (Questions 21–40): Solving the model and the Taylor principle

- 21) **Solving the two-equation system.** After deriving IS–MP, equilibrium inflation is solved by combining: _____
- (A) IS and MP only
 - (B) MP and money demand
 - (C) IS–MP and Phillips curve
 - (D) Phillips curve and money demand
- 22) **Solve for inflation (structure).** Solving yields a formula of the form: _____
- (A) $\pi_t = \theta\pi_t^e + (1 - \theta)\pi^* + \theta(\text{shocks})$
 - (B) $\pi_t = \pi_t^e + \pi^* + \theta(\text{shocks})$
 - (C) $\pi_t = \pi^*$ always
 - (D) $\pi_t = \theta\pi^* + (1 - \theta)\pi_t^e$
- 23) **Definition of θ .** In the Taylor principle notes: _____
- (A) $\theta = 1 + \alpha\gamma(\beta_\pi - 1)$
 - (B) $\theta = \frac{1}{1 + \alpha\gamma(\beta_\pi - 1)}$

(C) $\theta = \frac{1}{1 - \alpha\gamma(\beta_\pi - 1)}$

(D) $\theta = \alpha\gamma(\beta_\pi - 1)$

(24) **Case $\beta_\pi > 1$.** If $\beta_\pi > 1$ and $\alpha, \gamma > 0$, then: _____

(A) $\theta > 1$

(B) $\theta < 0$

(C) $0 < \theta < 1$

(D) $\theta = 1$

(25) **Parameter interpretation (review emphasis).** In $\theta = \frac{1}{1 + \alpha\gamma(\beta_\pi - 1)}$, θ is smaller when: _____

(A) α is smaller

(B) γ is smaller

(C) β_π is closer to 1

(D) α, γ , or β_π is larger (with $\beta_\pi > 1$)

(26) **Numerical θ .** Let $\alpha = 1, \gamma = 0.5, \beta_\pi = 3$. Then θ equals: _____

(A) $1/(1 + 0.5 \cdot 3) = 0.4$

(B) $1/(1 + 0.5 \cdot 2) = 0.5$

(C) $1/(1 + 0.5 \cdot 1) = 0.667$

(D) $1/(1 - 0.5 \cdot 2) = \infty$

(27) **Effect of tougher policy on inflation.** With $\beta_\pi > 1$, raising β_π tends to: _____

(A) reduce θ and make inflation closer to the target

(B) increase θ and make inflation closer to expectations

(C) reduce γ

(D) shift the Phillips curve down permanently

(28) **Adaptive expectations.** Under adaptive expectations: _____

- (A) $\pi_t^e = \pi^*$
- (B) $\pi_t^e = \pi_{t-1}$
- (C) $\pi_t^e = \mathbb{E}_t \pi_{t+1}$
- (D) $\pi_t^e = 0$ always

(29) **Accelerationist form.** With adaptive expectations, the Phillips curve becomes: _____

- (A) $\pi_t = \pi^* + \gamma(y_t - y_t^*) + \varepsilon_t^\pi$
- (B) $\pi_t = \pi_t^e - \gamma(y_t - y_t^*) + \varepsilon_t^\pi$
- (C) $\pi_t = \pi_{t-1} + \gamma(y_t - y_t^*) + \varepsilon_t^\pi$
- (D) $\pi_t = \pi_{t-1} - \gamma(y_t - y_t^*) + \varepsilon_t^\pi$

(30) **Interpretation.** Under adaptive expectations, there is a positive relationship between: _____

- (A) the change in inflation ($\pi_t - \pi_{t-1}$) and the output gap
- (B) the level of inflation and the output gap only
- (C) the output gap and the inflation target only
- (D) money growth and the output gap only

(31) **Stability under Taylor principle.** If $0 < \theta < 1$, then inflation dynamics $\pi_t = \theta\pi_{t-1} + \dots$ are: _____

- (A) explosive
- (B) oscillatory with increasing amplitude
- (C) a random walk
- (D) stable AR(1) mean reversion

- ③② **When $\beta_\pi < 1$.** If $\beta_\pi < 1$ (so policy violates Taylor principle), then inflation shocks tend to: _____
- (A) raise the real rate and reduce output, stabilizing inflation
 - (B) lower the real rate and raise output, amplifying inflation
 - (C) not affect the real rate
 - (D) stabilize inflation faster
- ③③ **Boundary emphasized in TaylorRule notes.** The interval producing $\theta > 1$ is: _____
- (A) $\left(1 - \frac{1}{\alpha\gamma}\right) < \beta_\pi < 1$
 - (B) $\beta_\pi > 1$
 - (C) $\beta_\pi < \left(1 - \frac{1}{\alpha\gamma}\right)$
 - (D) $\beta_\pi = 1$
- ③④ **Numerical boundary.** If $\alpha = 1$ and $\gamma = 0.5$, then $1 - \frac{1}{\alpha\gamma}$ equals: _____
- (A) 0
 - (B) 0.5
 - (C) -1
 - (D) 2
- ③⑤ **Graph logic (review emphasis).** When $\beta_\pi < 1$, the IS–MP curve in (y, π) space: _____
- (A) slopes down
 - (B) slopes up
 - (C) is horizontal
 - (D) is vertical

- 36) Condition “upward and steeper than PC”. The TaylorRule notes state that IS–MP slopes up and is steeper than PC when: _____
- (A) $\beta_\pi > 1$
 - (B) $\beta_\pi < 1 - \frac{1}{\alpha\gamma}$
 - (C) $\beta_\pi = 1$
 - (D) $1 - \frac{1}{\alpha\gamma} < \beta_\pi < 1$
- 37) Explosive dynamics intuition. In the explosive region $1 - \frac{1}{\alpha\gamma} < \beta_\pi < 1$, an increase in expected inflation π_t^e tends to: _____
- (A) start a sequence where inflation and output move further away from target over time
 - (B) reduce inflation and raise output back to natural values
 - (C) have no effect because expectations do not matter
 - (D) immediately pin inflation at the target
- 38) A puzzling result (TaylorRule slide 11). When $\beta_\pi < 1$, increasing the inflation target π^* can lead to: _____
- (A) higher inflation and higher output always
 - (B) no change in nominal interest rates
 - (C) higher interest rates and lower inflation (counterintuitive outcome)
 - (D) lower interest rates and higher inflation
- 39) Quick computation (IS–MP coefficient). If $\alpha = 1.5$ and $\beta_\pi = 0.8$, then $-\alpha(\beta_\pi - 1)$ equals: _____
- (A) -0.3
 - (B) $+0.3$
 - (C) -1.2
 - (D) $+1.2$

④⑩ **Policy interpretation.** If $-\alpha(\beta_\pi - 1) > 0$, then higher inflation is associated with:

- (A) higher output along IS–MP (destabilizing feedback)
- (B) lower output along IS–MP (stabilizing feedback)
- (C) no change in output
- (D) lower inflation